Introduction

Researchers have noted that sleep is essential for good health across multiple domains (i.e. education, clinical practice, health care) and, as a result, have called for more research into the effects of insufficient sleep (Ramar et al., 2021). Getting adequate sleep is important for cognition, mood, mental and physical health (Watson et al., 2015). Even short-term sleep deprivation can have a detrimental effect on physical and mental health, as well as public safety (Luyster et al., 2012). Our project explores the effects of alcohol and caffeine use on sleep efficiency and deep sleep. Sleep efficiency and deep sleep have long been used by researchers to measure sleep quality and sleep disorders (Reed & Sacco, 2016; Riedel & Lichstein, 1998). Alcohol and caffeine are the most commonly used depressants and stimulants, respectively (Costardi et al., 2015; Evans et al., 2025). Other variables used in our analysis include exercise, which has been shown to improve sleep quality (Alnawwar et al., 2023), and tobacco use, which, combined with caffeine and/or alcohol, have increased physiological impact (i.e. on blood pressure and anxiety) (Vinader-Caerols et al., 2012). Our analysis revealed substance use has similar and differing effects for males and females, with stronger effects for those who smoke compared to those who do not smoke.

Factoids:

About 85% of the U.S. population consumes at least one caffeinated beverage per day, containing about 165 mg. of caffeine (Mitchell et al., 2014).

As of February, 2025, about 79% of the U.S. population ages 12 and older reported they drank alcohol at some point in their lifetime (National Institute on Alcohol Abuse and Alcoholism, 2025)

Written observations next page.

Table 1: Correlations between sleep quality measures, substance use, and exercise in the combine sample							
	1	2	3	4	5	6	7
1. Sleep Efficiency	–	0.18*	0.81***	-0.59***	-0.13	-0.43***	0.34***
2. REM Sleep Percent	-0.06	–	-0.01	-0.06	0.15*	-0.14*	0.06
3. Deep Sleep Percent	0.77***	-0.37***	–	-0.36***	-0.18***	-0.44***	0.22***
4. Awakenings	-0.55***	0.03	-0.27***	–	0.07	0.24***	-0.33***
5. Caffeine Consumption	0.21**	-0.04	0.15*	-0.19**	–	0.11	-0.15*
6. Alcohol Consumption	-0.36***	0.04	-0.30***	0.17*	-0.24***	–	-0.01
7. Exercise Frequency	0.16*	0.08	0.10	-0.14*	0.12	0.00	–
$*\ p \leq 0.05$, $**\ p \leq 0.01$, $***\ p \leq 0.001$							
Note: Female participants above the diagonal, male participants below.							

Table 2: Correlations between sleep quality measures, substance use, and exercise among non-smokers							
	1	2	3	4	5	6	7
1. Sleep Efficiency	–	0.26***	0.72***	-0.63***	-0.07	-0.22**	0.43***
2. REM Sleep Percent	-0.07	–	0.00	-0.08	0.11	-0.12	0.11
3. Deep Sleep Percent	0.61***	-0.43***	–	-0.27***	-0.15	-0.30***	0.28***
4. Awakenings	-0.65***	0.02	-0.22*	–	0.10	0.15	-0.37***
5. Caffeine Consumption	0.17	-0.11	0.12	-0.20*	–	0.13	-0.20*
6. Alcohol Consumption	-0.27**	0.18*	-0.20*	0.17	-0.26***	–	-0.03
7. Exercise Frequency	0.31***	0.07	0.20*	-0.17	0.19*	-0.01	–
$*\ p \leq 0.05$, $**\ p \leq 0.01$, $***\ p \leq 0.001$							
Note: Female participants above the diagonal, male participants below.							

Table 1: Correlations between sleep quality measures, substance use, and exercise among smokers							
	1	2	3	4	5	6	7
1. Sleep Efficiency	–	0.13	0.91***	-0.59***	0.06	-0.60***	-0.02
2. REM Sleep Percent	-0.01	–	-0.01	0.01	0.20	-0.25	-0.11
3. Deep Sleep Percent	0.86***	-0.30**	–	-0.56***	-0.01	-0.57***	-0.08
4. Awakenings	-0.58***	0.07	-0.43***	–	-0.11	0.41**	-0.16
5. Caffeine Consumption	0.33**	0.19	0.17	-0.26*	–	-0.19	0.41**
6. Alcohol Consumption	-0.54***	-0.17	-0.48***	0.15	-0.29**	–	0.06
7. Exercise Frequency	0.03	0.10	-0.01	-0.07	-0.09	0.03	–
$*\ p \leq 0.05$, $**\ p \leq 0.01$, $***\ p \leq 0.001$							
Note: Female participants above the diagonal, male participants below.							

Combined population observations of interest:

Sleep efficiency

- Caffeine consumption and sleep efficiency are positively associated for males, neg. and not sig. for females.
- Alcohol consumption and sleep efficiency are negatively associated for both, slightly stronger for females.
- Exercise frequency and sleep efficiency are positively associated for both, twice as strong for females.

REM Sleep

- Caffeine consumption and REM sleep are positively associated for females, neg. and not sig. for males.
- Alcohol consumption and REM sleep are negatively associated for females, pos. and not sig. for males.

Deep Sleep

- Caffeine consumption and deep sleep are positively associated for males, negatively associated for females.
- Exercise frequency and deep sleep are positively associated for females, pos. and not sig. for males.

Awakenings

- Caffeine consumption and awakenings are negatively associated for males, pos. and not sig. for females.
- Alcohol consumption and awakenings are positively associated for both, slightly stronger for females.
- Exercise frequency and awakenings are negatively associated for both, much stronger for females.

Smoker population observations of interest:

Sleep efficiency

- Caffeine consumption and sleep efficiency are positively associated for males, pos. and not sig. for females.
- Alcohol consumption and sleep efficiency are negatively associated for both.

REM sleep

- No significant relationships.

Deep sleep

- Alcohol consumption and deep sleep are negatively associated for both.

Smoker population observations of interest (continued):

Awakenings

- Caffeine consumption and awakenings are negatively associated for males, neg. and not sig. for females.
- Alcohol consumption and awakenings are positively associated for females, pos. and not sig. for males.

Non-smoker population observations of interest:

Sleep efficiency

- Alcohol consumption and sleep efficiency are negatively associated for both.
- Exercise and sleep efficiency are positively associated for both, slightly stronger for females.

REM sleep

- Alcohol consumption and REM sleep are positively associated for males, neg. and not sig. for females.

Deep sleep

- Alcohol consumption and deep sleep are negatively associated for both, slightly stronger for females.
- Exercise frequency and deep sleep are positively associated for both, slightly stronger for females.

Awakenings

- Caffeine consumption and awakenings are negatively associated for males, pos. and not sig. for females.
- Exercise frequency and awakenings are negatively associated for females, neg. and not sig. for males.

References

Alnawwar, M. A., Alraddadi, M. I., Algethmi, R. A., Salem, G. A., Salem, M. A., & Alharbi, A. A. (2023). The effect of physical activity on sleep quality and sleep disorder: A systematic review. *Cureus*, *15*(8), 11. https://doi.org/10.7759/cureus.43595

Costardi, J. V. V., Nampo, R. A. T., Silva, G. L., Ribeiro, M. A. F., Stella, H. J., Stella, M. B., & Malheiros, S. V. P. (2015). A review on alcohol: From the central action mechanism to chemical dependency. *Revista Da Associação Médica Brasileira*, *61*(4), 381–387. https://doi.org/10.1590/1806-9282.61.04.381

Evans, J., Richards, J. R., & Battisti, A. S. (2025). Caffeine. In *StatPearls*. StatPearls Publishing. http://www.ncbi.nlm.nih.gov/books/NBK519490/

Luyster, F. S., Strollo, P. J., Jr., Zee, P. C., & Walsh, J. K. (2012). Sleep: A health imperative. *Sleep*, *35*(6), 727–734. https://doi.org/10.5665/sleep.1846

Mitchell, D. C., Knight, C. A., Hockenberry, J., Teplansky, R., & Hartman, T. J. (2014). Beverage caffeine intakes in the U.S. *Food and Chemical Toxicology: An International Journal Published for the British Industrial Biological Research Association*, *63*, 136–142. https://doi.org/10.1016/j.fct.2013.10.042

National Institute on Alcohol Abuse and Alcoholism. (2025, February). *Alcohol Use in the United States: Age Groups and Demographic Characteristics*. https://www.niaaa.nih.gov/alcohols-effects-health/alcohol-topics-z/alcohol-facts-and-statistics/alcohol-use-united-states-age-groups-and-demographic-characteristics

Ramar, K., Malhotra, R. K., Carden, K. A., Martin, J. L., Abbasi-Feinberg, F., Aurora, R. N., Kapur, V. K., Olson, E. J., Rosen, C. L., Rowley, J. A., Shelgikar, A. V., & Trotti, L. M. (2021). Sleep is essential to health: An American Academy of Sleep Medicine position statement. *Journal of Clinical Sleep Medicine*. https://doi.org/10.5664/jcsm.9476

Reed, D. L., & Sacco, W. P. (2016). Measuring sleep efficiency: What should the denominator be? *Journal of Clinical Sleep Medicine : JCSM : Official Publication of the American Academy of Sleep Medicine*, *12*(2), 263–266. https://doi.org/10.5664/jcsm.5498

Riedel, B. W., & Lichstein, K. L. (1998). Objective sleep measures and subjective sleep satisfaction: How do older adults with insomnia define a good night's sleep? *Psychology and Aging*, *13*(1), 159–163.

Vinader-Caerols, C., Monleón, S., Carrasco, C., & Parra, A. (2012). Effects of alcohol, coffee, and tobacco, alone or in combination, on physiological parameters and anxiety in a young population. *Journal of Caffeine Research*, *2*(2), 70–76. https://doi.org/10.1089/jcr.2012.0018

Watson, N. F., Badr, M. S., Belenky, G., Bliwise, D. L., Buxton, O. M., Buysse, D., Dinges, D. F., Gangwisch, J., Grandner, M. A., Kushida, C., Malhotra, R. K., Martin, J. L., Patel, S. R., Quan, S. F., & Tasali, E. (2015). Joint consensus statement of the American Academy of Sleep Medicine and Sleep Research Society on the recommended amount of sleep for a healthy adult: Methodology and discussion. *Journal of Clinical Sleep Medicine*. https://doi.org/10.5664/jcsm.4950